

AUTONOMOUS OPTIONS WHEEL · AI RISK GOVERNOR

# Trades on rails. AI on a **leash**.

A deterministic engine sells cash-secured puts and covered calls on hard-coded rails; a Claude layer above it holds exactly two powers — read the regime and veto a new entry. It can never place, size, or force a trade, and it can never block an exit.

**+54.6%**

selection-first **variant** — 2.5-yr backtest, bot picks 3 stocks/week — vs SPY +45.6%

**+32.2%**

the live **5-name engine**'s own 2.5-yr run — live & unattended on Alpaca paper since day 1

**100% public**

decision journal, backtest artifacts, bugs found & fixed in the open

## THE PROBLEM

# Options income is a discipline problem disguised as a trading problem

Retail premium sellers must run a boring, repeatable process — the wheel — and can't: the wrong delta, panic buy-backs, one oversized ticker. The usual "AI trading agent" answer makes it worse, because an LLM asked to **pick trades** hallucinates, drifts, and cannot be held to a risk mandate.

### MANUAL & EMOTIONAL

Discipline breaks exactly when it matters — challenged puts get bought back in fear, winners get oversized in greed.

### LLM PICKERS ARE UNSAFE

A model that can trade can churn an account for engagement and invent entries — no way to prove alignment.

### THE AUDIENCE IS HUGE

US options volume: **15.2B contracts in 2025**, a sixth straight record, with retail at 30–60% of flow.

## THE SOLUTION

# Invert the design: the AI governs, the engine trades

A **deterministic wheel engine** does 100% of the trading on hard-coded rails. A **Claude layer** sits above it as a risk governor with exactly two powers — and every refusal is journaled verbatim.

### ⚙️ **Deterministic engine — does everything**

- CSP at delta 0.30, DTE 7–35, premium  $\geq 0.5\%$  of collateral
- Take-profit at 50% · roll only for credit at delta 0.60
- Assignment → covered calls above  $\max(\text{cost}, \text{upper Bollinger})$
- Marketable-limit orders only, exits always before entries

### 🧠 **Claude governor — may only refuse**

- **Reads the regime** each cycle (SPY/QQQ trend, vol percentile, GLD/VIXY/BTC, filtered news) → RISK\_ON / NEUTRAL / RISK\_OFF
- **Vetoes new entries** — one refusal with a reason kills it
- Can never place, size, or force a trade — never block an exit or buy-back

Incentive alignment by construction: an AI that cannot trade cannot churn your account. A deterministic SPY anchor backs it up — the tighter of the two readers governs.

## HOW IT WORKS

# One cycle, six steps — every 5–15 minutes while the market is open

The broker is the source of truth at every step, and everything the agent decides is appended to a public JSONL decision journal.



### The regime scales exposure

RISK\_ON → 72% of equity as collateral · NEUTRAL → 36% · RISK\_OFF  
→ no new entries. Claude's judgment and a deterministic SPY-intraday  
anchor both apply — the tighter one wins.

### The journal is the product

Equity per cycle, every order with its reason, every regime read with its  
explanation, every error. On night 1 the journal itself caught two  
production bugs — both fixed live, in public.

## SAFETY

# Risk is hard-coded, not vibes

Every gate below is a deterministic check that runs before any order — the same gates are stress-tested live on the dashboard's Risk Lab tab.

### CONCENTRATION

**5 · 2 ·  
18%**

max underlyings ·  
max per correlated  
sector · max equity  
per name

### TOTAL COLLATERAL

**72% / 36%**

of equity as collateral  
(RISK\_ON /  
NEUTRAL) — the  
rest is dry powder

### DAILY DRAWDOWN STOP

**-3%**

no new entries after  
a 3% intraday equity  
drawdown

### MARKET SHOCK ANCHOR

**-2% ·  
-4%**

SPY intraday tiers:  
budget halves, then  
entries stop entirely

### EARNINGS BLACKOUT

**per-name**

no new entries near  
earnings dates  
(config-driven)

### KILL SWITCH

**1 file**

touch agent/KILL  
and the agent stops  
submitting anything

Plus: paper endpoint hard-coded (the code cannot reach live money), quality floors on every entry, exit rules that are never vetoed — de-risking is always allowed.

## LIVE RUN

# Live and unattended on Alpaca paper since day 1

The full system — engine, governor, gates, journal — has been trading paper money on its own from the first evening of the hackathon. Its own decision journal caught two production bugs on night 1; both were fixed live and committed in the open.

### PREMIUM COLLECTED

**\$1,753**

4 fills, net of marks

### AUTONOMOUS CYCLES

**25 live**

39 total incl. dry-runs

### REGIME READS

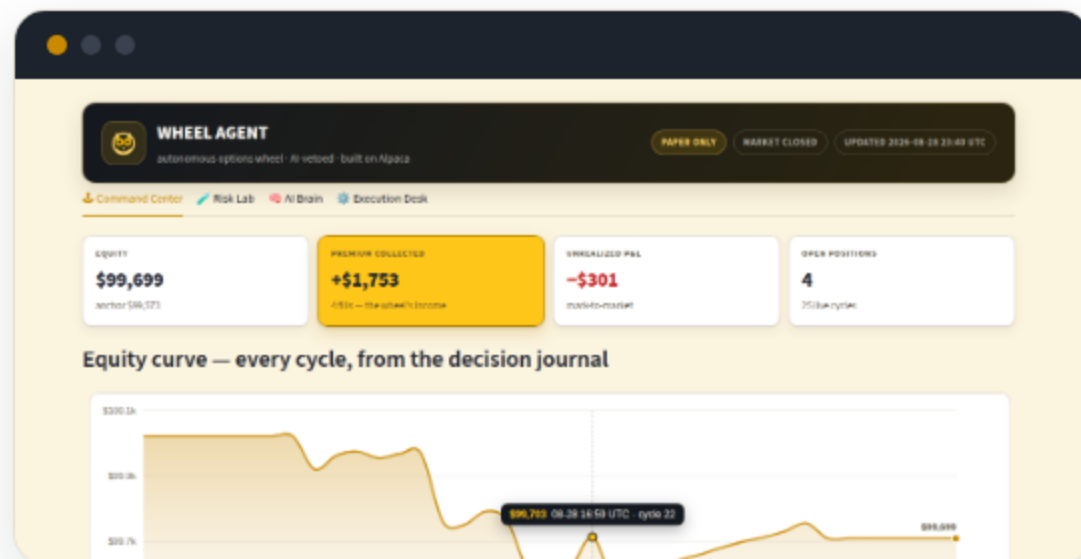
**31 / 8 / 0**

RISK\_ON / NEUTRAL / RISK\_OFF

### AI VETOES

**0**

path exercised in dry-run — none needed yet



The 6-tab dashboard: Command Center, Risk Lab (live stress test), AI Brain (regime reads & vetoes), Execution Desk, About (backtest methodology), FAQ.

## EVIDENCE

# Backtested 2.5 years on real option trades — beats SPY in its own bull window

The selection-first mode: each Monday the bot scores a 24-name universe (SMA200 quality gate, 63-day momentum, premium richness) and sells puts on the **top 3 names only**. Every premium is a **real Alpaca option-trade bar**; no parameter was fitted to this window.



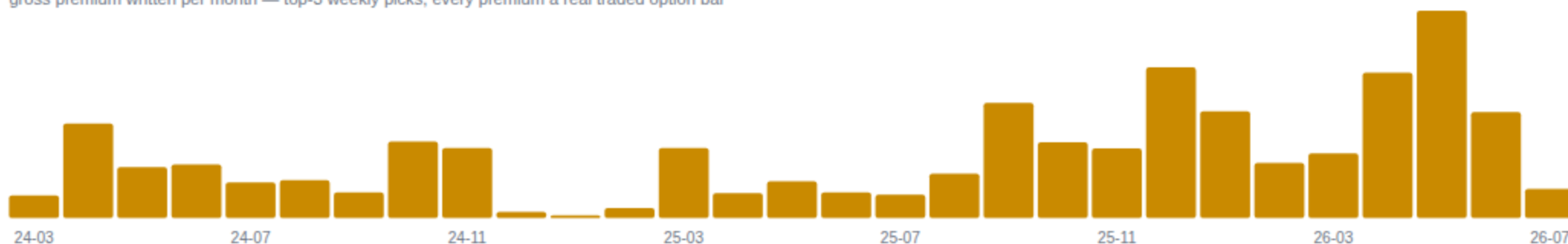
# Four configurations, one window, every result published

Each exploration fixes its rules **before** looking at out-of-sample data. We ran a series of backtests to beat SPY buy-and-hold — and picked the top-3 mode because it's **the only configuration that beat SPY in BOTH windows**, the 1-year test and the full 2.5 years.

| CONFIGURATION                                                          | WINDOW        | RETURN        | MAX DD       | VS SPY +45.6%                       |
|------------------------------------------------------------------------|---------------|---------------|--------------|-------------------------------------|
| <b>Live engine</b> — same rules as production                          | 2.5 yr        | +32.2%        | 18.3%        | trades upside for drawdown control  |
| <b>+ Defensive levers</b> — quality screen · regime delta · cash yield | 2.5 yr        | +34.2%        | <b>12.9%</b> | Calmar 1.51 vs 1.34 ✓               |
| <b>+ Weekly champion</b> — 1 pick/wk, 40% position                     | 1 yr → 2.5 yr | +46% → +35.3% | → 27.2%      | edge didn't survive the full window |
| <b>+ Top-3 weekly picks</b> — the slide-7 headline                     | 2.5 yr        | <b>+54.6%</b> | 21.5%        | beats SPY on return and Calmar ✓    |

The selection layer adds measurable value: names the bot picked averaged **+117%** vs the universe's +64%. K=1 beat SPY at 1 year (+46% vs +20%) but failed at 2.5 — rejected; K=2/K=5 lost outright; **K=3 won both windows** (+27% vs +20% at 1 yr) — selected. Momentum screens buy tops in mean-reverting years — which is why the failure row stays on this slide.

gross premium written per month — top-3 weekly picks, every premium a real traded option bar



BUSINESS CASE

# A paid governor in a market that has never been larger

US listed options printed **15.2B contracts in 2025** — the sixth consecutive record — with retail estimated at 30–60% of flow. Every competitor automates *execution*; none sells an AI risk governor whose alignment is provable.

TAM

**\$4–18B/yr**

~10–15M active US options traders at \$30–100/mo (Fermi, sources in repo)

SAM

**\$0.2–1.8B/yr**

systematic premium sellers who will connect a broker API

SOM · 3 YR

**\$1.5–12M ARR**

5–20k subscribers at \$25–50/mo, solo → small team

| COMPETITOR     | PRICE         | THEY AUTOMATE         | THE GAP WE FILL                                                                                                                                          |
|----------------|---------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Option Alpha   | \$99–149/mo   | your rules, executed  | No market judgment, no AI governor, no public journal. We bring the wheel determinism + a veto-only Claude layer + bring-your-own-broker (Alpaca first). |
| Composer       | \$32–40/mo    | stock algos           |                                                                                                                                                          |
| Option Samurai | \$39–49/mo    | finding trades        |                                                                                                                                                          |
| tastytrade     | \$0 + \$1/ctr | education & brokerage |                                                                                                                                                          |

**Revenue:** \$29/mo core (one broker, full engine + governor + dashboard) · \$79/mo pro (multi-account, spreads sleeve) — no payment-for-order-flow; an AI that can only say NO is also a pricing story.  
Sources: Cboe State of the Options Industry 2025 · MEMX · Devexperts · competitor pricing pages (Aug 2026)

## WHAT'S NEXT

# Built on all three Alpaca surfaces, with a public paper trail

### TRADING API

## alpaca-py

the engine: orders, OPRA greeks, buying-power  
clamps

### CLI

## alpaca 0.0.13

ops & monitoring · doctor before every session ·  
backtest data

### MCP SERVER

## connected

the broker surface, live inside the AI session that  
supervises it

**Next:** a defined-risk spreads sleeve, walk-forward re-validation of every gate, the weekly-champion selection mode behind an opt-in with tighter drawdown limits — and the veto-only governor offered as a safety layer for anyone else's trading agent.

 [github.com/guntoken/alpaca-wheel-agent](https://github.com/guntoken/alpaca-wheel-agent)

 [agent/journal.jsonl](#) — the decision journal

 [agent/runs/](#) — backtests with full artifacts

 [alpaca-wheel-agent.streamlit.app](https://alpaca-wheel-agent.streamlit.app)

Paper trading only · options involve substantial risk · hypothetical backtests don't guarantee future results · not investment advice.